hipDec:1.7

API Reference

X Platform, IP JPEG XS Decoder 2110 1.16.1

09/07/2025

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1 Overview

Changelog:

1.7

- **Changed**
  - The imported package `nmosConfig` is bumped from 1.0 to 1.1.
  - The imported package `hipCardSettings` is bumped from 1.2 to 1.3.

1.6

- **Added**
  - `nmosStatus` 1.1.

1.5

- **Changed**
  - `hipDecoder` from 1.3 to 1.4.
  - `hipDecDeskew` from 1.1 to 1.2.
  - `hipDecStatus` from 1.3 to 1.4.
  - `hipDecTransport` from 1.3 to 1.4.
  - `hipStatus` from 1.3 to 1.4.
  - `hipTypes` from 1.3 to 1.4.
  - `hipIpInterface` from 1.0 to 1.2.
  - `hipNetworkStatus` from 1.2 to 1.3.
  - `hipPhysicalPorts` from 1.0 to 1.1.
- **Added**
  - Added `linkModes` 1.3.
  - Added `lldp` 1.1.
  - Added `networkManagerProtocols` 1.1.
  - Added `lldpProtocols` 1.0.

1.4

- **Changed**
  - `hipDecoder` from 1.2 to 1.3.
  - `hipDecDeskew` from 1.0 to 1.1.
  - `hipDecTransport` from 1.2 to 1.3.
  - `hipStatus` from 1.2 to 1.3.
  - `hipTypes` from 1.2 to 1.2.

1.3

- **Added**
  - `hipDecDeskew` 1.0.
  - `hipIpInterface` 1.0.

  - `hipNetworkStatus` 1.0.
  - `hipPhysicalPorts` 1.0.
  - `hipCardSettings` 1.0.
  - `physicalIpPort` 1.0.
  - `sfpConstants` 1.0.
  - `sfpStatus` 1.0.
- **Changed**
  - `hipDecStatus` from 1.1 to 1.2.

1.2

- **Added**
  - `hipTestGenerator` 1.0.
- **Changed**
  - `hipDecoder` from 1.1 to 1.2.
  - `hipDecStatus` from 1.0 to 1.1.
  - `hipDecTransport` from 1.1 to 1.2.
  - `hipStatus` from 1.1 to 1.2.
  - `hipTypes` from 1.1 to 1.2.

1.1

- **Added**
  - `hipDecStatus` 1.0.
- **Changed**
  - `hipDecoder` from 1.0 to 1.1.
  - `hipDecTransport` from 1.0 to 1.1.
  - `hipStatus` from 1.0 to 1.1.
  - `hipTypes` from 1.0 to 1.1.

2 commonTypes (1.0)

2.1 Type Reference

2.1.1 NmosVersionNumber

Generic version number type for NMOS APIs

enum

V1_2	
V1_3	

3 firewallTypes (1.0)

3.1 Type Reference

3.1.1 IpAddress

string

3.1.2 PaginatedRequest

```
Instructs the server to return only a subset of the data found on the server

@param pageSize number of entries returned per request
@param pageNumber what page of the set to returned, determined by the slice [pageSize*pageNumber
```

struct

pageSize	int
pageNumber	int

3.1.3 PaginationInfo

struct

totalEntries	int the total number of entries on the server, before matching

3.1.4 SocketAddress

Tuple with IP address and port

struct

address	IpAddress IP address, either IPv4 or IPv6
port	int

3.1.5 UUID

string

4 hipCardSettings (1.3)

4.1 Overview

Changelog:

1.3

- **Changed**
  - The imported package `nmosConfig` is bumped from 1.0 to 1.1.

1.2

- **Added**
  - The value nmosConfig is added to cardsettings.

4.2 Command Reference

4.2.1 GetCardSettings

- **message GetCardSettings.Request**
- **message GetCardSettings.Response**

4.2.2 SetCardSettings

CAUTION: Always use the same UUID as the object already existing on the card.

- **message SetCardSettings.Request**
- **message SetCardSettings.Response**

4.3 Type Reference

4.3.1 CardSettings

- Settings for the operational mode of the card.
-
-

struct

`clockMode`	**ClockMode** Type of clock synchronization protocol the card should use •
`nmos`	**optional nmosConfig.NmosConfig** Parameters for accessing NMOS registry and labeling templates

4.3.2 ClockMode

Different types of clock synchronization protocols

enum

PTP	Sets the card to Precision Time Protocol (IEEE 1588)(PTP) time synchronization
LOCAL	Sets the card to Network Time Protocol (NTP) time synchro-nization or local clock if the card does not get a NTP lock.

4.3.3 GetCardSettings.Request

empty **struct**

4.3.4 GetCardSettings.Response

struct

data	**map** from **UUID** to **CardSettings**

4.3.5 SetCardSettings.Request

struct

data	**map** from **UUID** to **CardSettings**

4.3.6 SetCardSettings.Response

empty **struct**

5 hipDecDeskew (1.2)

5.1 Overview

Changelog:

1.2

- **Changed**
  - `hipTypes` from 1.3 to 1.4;

1.1

- **Changed**
  - `hipTypes` from 1.2 to 1.3;

5.2 Command Reference

5.2.1 ApplyDeskew

Applies deskew to all essences for Decoder objects

NOTE: this will create a break in transmission. Request: map from decoder uuid -> maxDelay

If the card has ptp lock: sets maxDelay to the supplied value for each group and enables ptp (dynamic) deskew. Else: measures the current skew between all present essences and applies a static deskew to them, taking into account the supplied maximum delay.

- **message ApplyDeskew.Request**
- **message ApplyDeskew.Response**

5.2.2 GetAppliedDeskew

Getter for applied deskew values

Request: list over decoder uuids Response: optional(Deskew), returning the applied deskew for the decoders that have deskew applied.

- **message GetAppliedDeskew.Request**
- **message GetAppliedDeskew.Response**

5.2.3 ClearDeskew

Clears deskew for given decoders

Request: list over decoder uuids to remove deskew for

- **message ClearDeskew.Request**
- **message ClearDeskew.Response**

5.3 Type Reference

5.3.1 AppliedDeskew

Type of deskew applied to group

variant

`measured`	**MeasuredDeskew** Measured deskew applied (no ptp lock when deskew was applied)
`ptp`	**PtpDeskew** PTP based deskew applied

5.3.2 ApplyDeskew.Request

struct

`data`	**map** from **UUID** to **float**

5.3.3 ApplyDeskew.Response

empty **struct**

5.3.4 ClearDeskew.Request

struct

`data`	**list** of **UUID**

5.3.5 ClearDeskew.Response

empty **struct**

5.3.6 Deskew

Deskew settings for group

struct

`maxDelay`	**float** Maximum delay in deskew path (ms)
`applied`	**AppliedDeskew** Deskew type applied

5.3.7 GetAppliedDeskew.Request

struct

`data`	**list** of **UUID**

5.3.8 GetAppliedDeskew.Response

struct

`data`	**map** from **UUID** to **optional** **Deskew**

5.3.9 MeasuredDeskew

Measured deskew applied to group

struct

`video`	**optional float** Deskew of the video essence (ms)
`audioMap`	**map** from **hipTypes.AudioIndex** to **float** Map of deskew of the audio essences (ms)
`anc`	**optional float** Deskew of the ancillary essence (ms)

5.3.10 PtpDeskew

PTP adjusted deskew (parameterless)

empty **struct**

6 hipDecStatus (1.4)

6.1 Overview

Changelog:

1.4

- **Changed**
- `hipTypes` from 1.3 to 1.4;

1.3

- **Changed**
- `hipTypes` from 1.2 to 1.3;

1.2

- **Added**
  - The fields skew and correctedSkew are added to the type TransportStatus.

1.1

- **Added**
  - The command ClearAllCounters is added.
- **Changed**
  - The imported package `hipTypes` is bumped from 1.1 to 1.2.
  - The fields rtpSsrc and rtpPayloadType on type RTPStatus are changed from being int to being optional(int).
  - The field rtpStatus on type TransportStatus is no longer optional.

6.2 Command Reference

6.2.1 GetDecoderTransportStatus

- **message GetDecoderTransportStatus.Request**
- **message GetDecoderTransportStatus.Response**

6.2.2 ClearAllCounters

- **message ClearAllCounters.Request**
- **message ClearAllCounters.Response**

6.3 Type Reference

6.3.1 ClearAllCounters.Request

empty **struct**

6.3.2 ClearAllCounters.Response

empty **struct**

6.3.3 DecoderTransportStatus

Decoder Transport Status

Status is reported for all configured essences.

struct

`videoStatus`	**optional TransportStatus** Video essence status
`audioStatus`	**map** from **hipTypes.AudioIndex** to **TransportStatus** Map of audio essence status
`ancStatus`	**optional TransportStatus** Anc essence status

6.3.4 DecoderTransportStatusRequest

struct

`ids`	**optional list** of **UUID**

6.3.5 DecoderTransportStatusResponse

struct

`data`	**map** from **UUID** to **DecoderTransportStatus**

6.3.6 GetDecoderTransportStatus.Request

DecoderTransportStatusRequest

6.3.7 GetDecoderTransportStatus.Response

DecoderTransportStatusResponse

6.3.8 NetworkLatencyStatus

Network latency status

struct

`minDelay`	**float** Minimum timestamp difference (millisec)
`maxDelay`	**float** Maximum timestamp difference (millisec)

6.3.9 RTPStatus

RTP status

struct

`bitrate`	**float**
`packetsMissing`	**bigint**
`sequenceErrors`	**bigint**
`rtpPayloadType`	**optional int**
`rtpSsrc`	**optional int**

6.3.10 TransportStatus

Transport Status for a Single Path

struct

`rtpStatus`	**RTPStatus** RTP status
`networkLatencyStatus`	**optional NetworkLatencyStatus** Network latency status (empty if no bitrate)
`skew`	**optional float** Skew of the essence (ms)
`correctedSkew`	**optional float** Skew of the essence after correction (ms)

7 hipDecTransport (1.4)

7.1 Overview

Changelog:

1.4

- **Added**
  - Added rtpPayload to DestinationSettings.
- **Changed**
  - The imported package `hipTypes` is bumped from 1.3 to 1.4.

1.3

- **Changed**
  - The imported package `hipTypes` is bumped from 1.2 to 1.3.

1.2

- **Changed**
  - The imported package `hipTypes` is bumped from 1.1 to 1.2.

1.1

- **Changed**
  - The imported package `hipTypes` is bumped from 1.0 to 1.1.

7.2 Type Reference

7.2.1 DestinationSettings

Transport settings for an UDP output

struct

`network`	**NetworkParameters** Network parameters
`output`	**OutputUdp** UDP output - cloned or single port
`rtpPayload`	**int** RTP payload type

7.2.2 NetworkParameters

Network parameters for Output

struct

`dscp`	**int** Type of service (TOS in IP header) Also named Differentiated Services Code Point (DSCP)
`ttl`	**int** Time to live (TTL in IP header)

`vlanPriority`	**optional VlanPriority** Vlan priority. Can only be used if output is on a vlan interface

7.2.3 OutputUdp

UDP output - select cloned output or single output mode

variant

`single`	**OutputUdpSettings** Configuration for the only port used by this output
`cloned`	**OutputUdpCloned** Configuration for port A and port B

7.2.4 OutputUdpCloned

Cloned Configuration for port A and port B

Both interfaces must be configured to run at the same link speed.

struct

`sourceA`	**OutputUdpSettings** Settings on interface D1
`sourceB`	**OutputUdpSettings** Settings on interface D2

7.2.5 OutputUdpSettings

Output settings for UDP outputs

struct

`interfaceId`	**UUID** Interface used
`destination`	**hipTypes.SocketAddress** Unicast or multicast address
`sourceAddress`	**optional string** Overwrite source address in IGMPv3 header
`sourcePort`	**optional int** Overwrite source port in UDP header

7.2.6 VlanPriority

VlanPriority - Set priority related 802.1Q header flags

Value: Traffic types 0 Best effort 1 Background (lowest priority) 2 Excellent effort 3 Critical applications 4 Video, < 100 ms latency and jitter 5 Voice, < 10 ms latency and jitter 6 Internetwork control

struct

`pcp`	**int** Priority code point

`dei`	**bool** Drop eligleble indicator

8 hipDecoder (1.4)

8.1 Overview

Changelog:

1.4

- **Changed**
  - The imported package `hipTypes` is bumped from 1.3 to 1.4.
  - The imported package `hipDecTransport` is bumped from 1.3 to 1.4.

1.3

- **Changed**
  - The imported package `hipTypes` is bumped from 1.2 to 1.3.
  - The imported package `hipDecTransport` is bumped from 1.2 to 1.3.
  - The field `decodeVideo` in type `DecoderVideoEssence` is changed from boolean to optional of the enum CodingCore.

1.2

- **Added**
  - The imported package `hipTestGenerator` 1.0 is added.
  - The fields testGenerator and policing are added to DecoderVideoEssence.
- **Changed**
  - The imported package `hipDecTransport` is bumped from 1.1 to 1.2.
  - The imported package `hipTypes` is bumped from 1.1 to 1.2.
  - The field source is moved from types DecoderVideoEssence, DecoderAncEssence and DecoderAudioEssence to type DecoderEssenceSettings.

1.1

- **Changed**
  - The imported package `hipDecTransport` is bumped from 1.0 to 1.1.
  - The imported package `hipTypes` is bumped from 1.0 to 1.1.

8.2 Command Reference

8.2.1 GetDecoders

Get Decoder objects from database.

- **message GetDecoders.Request**
- **message GetDecoders.Response**

8.2.2 SetDecoders

Update Decoder objects in database.

- **message SetDecoders.Request**

- message **SetDecoders.Response**

8.2.3 DeleteDecoders

Delete Decoder objects from database.

- message **DeleteDecoders.Request**
- message **DeleteDecoders.Response**

8.3 Type Reference

8.3.1 Decoder

Decoder Config

Decoder configuration

struct

`enable`	**bool** Enable decoder
`label`	**string** Label of decoder
`source`	**DecoderSource2110** ST2110 source struct
`onSignalLoss`	**hipTypes.SignalLossAction** Action on signal loss

8.3.2 DecoderAncEssence

Decoder ST2110 Anc Essence

Specification for ST2110 ancillary essence source Source has to be of ST2110 ancillary essence type

struct

`settings`	**DecoderEssenceSettings** Essence settings

8.3.3 DecoderAudioEssence

Decoder ST2110 Audio Essence

Specification for ST2110 audio essence source Source has to be of ST2110 audio essence type

struct

`settings`	**DecoderEssenceSettings** Essence settings

8.3.4 DecoderEssenceSettings

Decoder ST2110 Essence Settings

Settings for ST2110 essence source

struct

`enable`	**bool** Enable essence
`source`	**UUID** Source uuid
`label`	**string** Label of essence
`destination`	**hipDecTransport.DestinationSettings** Transport settings

8.3.5 DecoderSource2110

Decoder ST2110 Source

Specification of ST2110 source

struct

`videoEssence`	**optional DecoderVideoEssence** Optional Video essence of source
`audioEssences`	**map** from **hipTypes.AudioIndex** to **DecoderAudioEssence** List over audio essences
`ancEssence`	**optional DecoderAncEssence** Optional ancillary essence

8.3.6 DecoderVideoEssence

Decoder ST2110 Video Essence

Specification for ST2110 video essence source Source has to be of ST2110 video essence type

struct

`settings`	**DecoderEssenceSettings** Essence settings
`decodeVideo`	**optional hipTypes.CodingCore** Decode (using JPEG XS) in UHD/HD Core or passthrough
`policing`	**optional hipTypes.VideoFormat** Video format policing rules
`testGenerator`	**hipTestGenerator.VideoTestGenerator** Video test generator settings

8.3.7 DeleteDecoders.Request

struct

`ids`	**list** of **UUID**

8.3.8 DeleteDecoders.Response

empty **struct**

8.3.9 GetDecoders.Request

empty **struct**

8.3.10 GetDecoders.Response

struct

`data`	**map** from **UUID** to **Decoder**

8.3.11 SetDecoders.Request

struct

`data`	**map** from **UUID** to **Decoder**

8.3.12 SetDecoders.Response

empty **struct**

9 hipIpInterface (1.2)

9.1 Overview

Changelog:

1.2

- **Changed**
  - The imported package `hipPhysicalPorts` is bumped from 1.0 to 1.1.

1.1

- **Added**
  - The `lldp` field is added to the `IpInterface` object.
  - The `GetLldpNeighbors` RPC is added

9.2 Command Reference

9.2.1 GetIpInterfaces

- **message GetIpInterfaces.Request**
- **message GetIpInterfaces.Response**

9.2.2 SetIpInterfaces

- **message SetIpInterfaces.Request**
- **message SetIpInterfaces.Response**

9.2.3 DeleteIpInterfaces

- **message DeleteIpInterfaces.Request**
- **message DeleteIpInterfaces.Response**

9.2.4 GetLldpNeighbors

Get LLDP Neighbors.

Returns data about the LLDP neighbor seen per interface. To see data for a specific port, LLDP must be enabled for that Interface.

- **message GetLldpNeighbors.Request**
- **message GetLldpNeighbors.Response**

9.3 Type Reference

9.3.1 DeleteIpInterfaces.Request

struct

`ids`	**list** of **UUID**

9.3.2 DeleteIpInterfaces.Response

empty **struct**

9.3.3 GetIpInterfaces.Request

empty **struct**

9.3.4 GetIpInterfaces.Response

struct

`data`	**map** from **UUID** to **IpInterface**

9.3.5 GetLldpNeighbors.Request

GetLldpNeighborsRequest

9.3.6 GetLldpNeighbors.Response

GetLldpNeighborsResponse

9.3.7 GetLldpNeighborsRequest

Get LLDP Neighbors Request

struct

`interface`	**optional list** of **UUID** List of UUIDs uniquely identifying IpInterfaces. Only LLDP Neighbors from the specified IpInterfaces are returned. If no value is set, the request will return neighbors from all ports that have LLDP enabled.
`details`	**bool** If false only chassisID, mgmtIP, and portID will be returned for each LLDP Neighbor. If true all available information is returned.

9.3.8 GetLldpNeighborsResponse

struct

`data`	**map** from **UUID** to **optional** **lldp.LldpNeighbor**

9.3.9 IpInterface

IP Interface

struct

`label`	**string** User label

`physicalPortName`	**physicalIpPort.PortName** The physical port used by the ip interface
`enabled`	**bool** Enable/disable the IP interface
`vlan`	**int** Vlan number. vlan 0 indicates no vlan in use.
`quad`	**optional QuadPortName** Interface number when physical port is set to quad mode (4x10G). Is optional.
`ipv4`	**optional IpInterfaceConfig** IPv4 interface configuration. Is optional.
`ipv6`	**optional IpInterfaceConfig** IPv4 interface configuration. Is optional.
`lldp`	**optional lldp.LldpMode** The LldpMode to set for the port. Leaving this empty indicates LLDP off.

9.3.10 IpInterfaceConfig

IP Interface configuration

struct

`ipAddress`	**string** IP address
`gateway`	**string** Gateway address of the interface
`netmask`	**int** Netmask of the interface in CIDR subnet mask notation e.g /24 = 255.255.255.0

9.3.11 QuadPortName

enum

`Q1`	
`Q2`	
`Q3`	
`Q4`	

9.3.12 SetIpInterfaces.Request

struct

`data`	**map** from **UUID** to **IpInterface**

9.3.13 SetIpInterfaces.Response

empty **struct**

10 hipNetworkStatus (1.3)

10.1 Overview

Changelog:

1.3

- Changed
  - The imported package `hipPhysicalPorts` is bumped from 1.0 to 1.1.

1.2

- Changed
  - The imported package `sfpStatus` is bumped from 1.3 to 1.4.

1.1

- Added
  - The imported package `sfpStatus` 1.3 is added.
  - The command GetSfpStatus is added.

10.2 Command Reference

10.2.1 GetNetworkStatus

- message GetNetworkStatus.Request
- message GetNetworkStatus.Response

10.2.2 GetSfpStatus

- message GetSfpStatus.Request
- message GetSfpStatus.Response

10.3 Type Reference

10.3.1 GetNetworkStatus.Request

NetworkStatusRequest

10.3.2 GetNetworkStatus.Response

NetworkStatusResponse

10.3.3 GetSfpStatus.Request

GetSfpStatusRequest

10.3.4 GetSfpStatus.Response

GetSfpStatusResponse

10.3.5 GetSfpStatusRequest

Get SFP Status Request

struct

physicalports	**optional list** of **UUID** List of UUIDs uniquely identifying PhysicalPorts. Only SFP Status from the specified PhysicalPorts are returned. If no value is set, the request will return SFP Status from all ports.

10.3.6 GetSfpStatusResponse

struct

data	**map** from **UUID** to **sfpStatus.SfpStatus**

10.3.7 LinkSpeed

LinkSpeed

enum

NO_LINK	No linkspeed
LINK_10_MBPS	10 Mbps
LINK_100_MBPS	100 Mbps
LINK_1_GBPS	1 Gbps
LINK_10_GBPS	10 Gbps
LINK_25_GBPS	25 Gbps
LINK_40_GBPS	40 Gbps

10.3.8 NetworkStatus

NetworkStatus

struct

macAddress	**string** Interface MAC address.
linkSpeed	**LinkSpeed** Linkspeed on interface.
rx	**optional int** Received bitrate on interface (kbps). Only applicable for the IP JPEG XS Encoder.
tx	**optional int** Transmitted bitrate on interface (kbps). Only applicable for the IP JPEG XS Decoder.

10.3.9 NetworkStatusRequest

NetworkStatusRequest

struct

`ids`	**optional list** of **UUID** Optional list of interface uuids

10.3.10 NetworkStatusResponse

NetworkStatusResponse

struct

`data`	**map** from **UUID** to **NetworkStatus** Map of networkstatus with interface uuid as key

11 hipPhysicalPorts (1.1)

11.1 Overview

Changelog:

1.1

- **Changed**
  - The `portMode` field in `PhysicalPort` was changed to `linkMode` and is using the `IpLinkMode` type from `linkModes` 1.3.
- **Added**
  - A `fecMode` field was added to the `PhysicalPort` object using the `IpFecMode` type from `linkModes` 1.3.

11.2 Command Reference

11.2.1 GetPhysicalPorts

- **message GetPhysicalPorts.Request**
- **message GetPhysicalPorts.Response**

11.2.2 SetPhysicalPorts

- **message SetPhysicalPorts.Request**
- **message SetPhysicalPorts.Response**

11.3 Type Reference

11.3.1 GetPhysicalPorts.Request

empty **struct**

11.3.2 GetPhysicalPorts.Response

GetPhysicalPortsResponse

11.3.3 GetPhysicalPortsResponse

struct

`data`	**map** from **UUID** to **PhysicalPort**

11.3.4 PhysicalPort

Physical Port

struct

`name`	**physicalIpPort.PortName** Name of the physical port

`label`	**string** User label
`linkMode`	**linkModes.IpLinkMode** Port mode determines link speed and quad mode
`fecMode`	**optional linkModes.IpFecMode** Optional FEC mode
`enabled`	**bool** Enable / disable the physical port

11.3.5 SetPhysicalPorts.Request

SetPhysicalPortsRequest

11.3.6 SetPhysicalPorts.Response

empty **struct**

11.3.7 SetPhysicalPortsRequest

struct

`data`	**map** from **UUID** to **PhysicalPort**

12 hipStatus (1.4)

12.1 Overview

Changelog:

1.4

- **Changed**
  - The imported package `hipTypes` is bumped from 1.3 to 1.4.

1.3

- **Changed**
  - The imported package `hipTypes` is bumped from 1.2 to 1.3.

1.2

- **Changed**
  - The imported package `hipTypes` is bumped from 1.1 to 1.2.
  - The field format on type VideoStatus is now optional.

1.1

- **Added**
  - The command GetThumbnailPath is added.
  - The field lineNumber is added to type DataIdentifier.
- **Changed**
  - The imported package `hipTypes` is bumped from 1.0 to 1.1.

12.2 Command Reference

12.2.1 GetEssenceStatus

Get Encoder essence status

- **message GetEssenceStatus.Request**
- **message GetEssenceStatus.Response**

12.2.2 GetThumbnailPath

Get Video thumbnail path

- **message GetThumbnailPath.Request**
- **message GetThumbnailPath.Response**

12.3 Type Reference

12.3.1 AncStatus

Ancillary Status

struct

`dataIdentifiers`	**list** of **DataIdentifier** List of data identifiers

12.3.2 AudioStatus

Audio Status

struct

`totalChannels`	**int** Total number of of Audio Channels
`standard`	**optional st2110Types.St2110Standard** St2110 audio standard
`packetTime`	**hipTypes.St2110AudioPacketTime** Packet time

12.3.3 DataIdentifier

Data Identifiers

The Data Identifier (DID) and Secondary Data Identifier (SDID) words which together indicate a type of ancillary data.

struct

`did`	**int** Data Identifier word
`sdid`	**int** Secondary Data Identifier word
`lineNumber`	**int** Line number of the received DID/SDID pair

12.3.4 EssenceStatus

Essence Status

Status is reported for all configured essences that have bitrate.

struct

`videoStatus`	**optional VideoStatus** Video status
`audioStatus`	**map** from **hipTypes.AudioIndex** to **AudioStatus** Map of audio statuses
`ancStatus`	**optional AncStatus** Ancillary status

12.3.5 EssenceStatusRequest

Essence Status Request

struct

`ids`	**optional list** of **UUID** Optional list of Encoder/Decoder UUIDs to request status for. If ids are empty, status for all UUIDs present will be returned.

12.3.6 EssenceStatusResponse

Essence Status Response

struct

`data`	**map** from **UUID** to **EssenceStatus** Map from Encoder/Decoder UUID to the related essence status structure.

12.3.7 EssenceThumbnailPathRequest

Essence Thumbnail Path Request

struct

`ids`	**optional list** of **UUID** Optional list of Encoder/Decoder UUIDs to request thumbnail path for. If ids are empty, status for all UUIDs present will be returned.

12.3.8 EssenceThumbnailPathResponse

Essence Thumbnail Path Response

struct

`data`	**map** from **UUID** to **string** Map from Encoder/Decoder UUID to the related essence thumbnail path.

12.3.9 GetEssenceStatus.Request

EssenceStatusRequest

12.3.10 GetEssenceStatus.Response

EssenceStatusResponse

12.3.11 GetThumbnailPath.Request

EssenceThumbnailPathRequest

12.3.12 GetThumbnailPath.Response

EssenceThumbnailPathResponse

12.3.13 VideoStatus

Video Status

struct

`compressed`	**bool** True if compressed, and false if raw
`format`	**optional hipTypes.VideoFormat** Video Resolution and Frame Rate

13 hipTestGenerator (1.0)

13.1 Type Reference

13.1.1 BitmapBackgroundProperty

Specifies the color property of the background in a bitmap.

enum

BLACK	
TRANSPARENT	

13.1.2 Color

A color defined in terms of RGB.

struct

red	int
green	int
blue	int

13.1.3 MovingBox

Specify the properties of a moving box to be used in a test image generator.

struct

enabled	bool
color	optional Color

13.1.4 TestPattern

Pattern to display through generator

enum

BLACK_PICTURE	Display black picture
COLOR_BARS	Display color bars

13.1.5 TextOverlay

Specify the properties of a text overlay to put on top of a test image.

Uses the label of the De/Encoder object as text to display. Text will be truncated after 36 characters

struct

enabled	bool
position	TextPosition
background	BitmapBackgroundProperty

13.1.6 TextPosition

Specify the vertical placement for the text.

enum

`TOP`	
`MIDDLE`	
`BOTTOM`	

13.1.7 VideoTestGenerator

Defines how to generate a test image for a specific port.

struct

`forceGenerator`	**bool**
`pattern`	**TestPattern**
`textOverlay`	**TextOverlay**
`movingBox`	**MovingBox**

14 hipTypes (1.4)

14.1 Overview

Changelog:

1.4

- **Added**
  - The enum StartChannel

1.3

- **Added**
  - The value R2160P to ResolutionScan

1.2

- **Removed**
  - The values BLACK_PICTURE, MOVING_BOX and COLOR_BARS are removed from type Signal-LossActionVideo.

1.1

- **Added**
  - The type TimeReference is added.

14.2 Type Reference

14.2.1 AudioIndex

Audio Index

enum

AUDIO_1	Audio index 1
AUDIO_2	Audio index 2
AUDIO_3	Audio index 3
AUDIO_4	Audio index 4
AUDIO_5	Audio index 5
AUDIO_6	Audio index 6
AUDIO_7	Audio index 7
AUDIO_8	Audio index 8

14.2.2 CodingCore

Coding Core Type

Type of Coding Core to use.

enum

UHD	Use Core that supports UHD/HD/SD
HD	Use Core that supports HD/SD

14.2.3 FrameRate

Frame Rate

Frame rate of the source.

enum

F24	24 fps
F23_98	23.98 fps
F25	25 fps
F29_97	29.97 fps
F30	30 fps
F47_95	47.95 fps
F48	48 fps
F50	50 fps
F59_94	59.94 fps
F60	60 fps

14.2.4 NumberOfChannels

Number of Channels

enum

CHANNELS_2	2 Mono Channels
CHANNELS_4	4 Mono Channels
CHANNELS_6	6 Mono Channels
CHANNELS_8	8 Mono Channels

14.2.5 Order

Order to put audio channel in.

enum

FIRST	Put audio channels first
SECOND	Put audio channels second
THIRD	Put audio channels third
FOURTH	Put audio channels fourth
FIFTH	Put audio channels fifth
SIXTH	Put audio channels sixth
SEVENTH	Put audio channels seventh
EIGHTH	Put audio channels eighth

14.2.6 ResolutionScan

Resolution and Scan

Resolution and scan mode of the source.

enum

R2160P	2096x2160 progressive

R2KP	2048x1080 progressive
R1080P	1920x1080 progressive
R1080I	1920x1080 interlaced
R720P	1280x720 progressive
R576I	720x576 interlaced
R480I	720x480 interlaced

14.2.7 SignalLossAction

Signal Loss Action

Action taken on loss of bitrate on essences.

struct

videoAction	**SignalLossActionVideo** Action on video essence
audioAction	**SignalLossActionAudio** Action on audi essences
ancAction	**SignalLossActionAnc** Action on ancillary essence

14.2.8 SignalLossActionAnc

Signal Loss Action on ancillary essence

Action taken on loss of bitrate on the ancillary essence.

enum

MUTE	Mutes ancillary essence
TEST_GEN	Generates test ancillary essence

14.2.9 SignalLossActionAudio

Signal Loss Action on audio essences

Action taken on loss of bitrate on the audio essences.

enum

MUTE	Mutes audio essence
TEST_GEN	Generates test audio

14.2.10 SignalLossActionVideo

Signal Loss Action on video essence

Action taken on loss of bitrate on the video essence.

enum

MUTE	Mutes video essence
TEST_GEN	Display picture from test generator

14.2.11 SocketAddress

Socket address

Tuple with IP address and port

struct

`address`	**string** IP address, either IPv4 or IPv6
`port`	**int** Port number

14.2.12 St2110AudioPacketTime

ST2110 Audio packet time

enum

`N_A`	No audio packet time
`US83_3`	83,3 us packet time
`US125`	125 us packet time
`MS1`	1 ms packet time

14.2.13 StartChannel

Channel to start on, starts at 1 up to 15 (Only even numbered channel pairs)

enum

`CHANNEL_1`	Starting at Channel 1
`CHANNEL_3`	Starting at Channel 3
`CHANNEL_5`	Starting at Channel 5
`CHANNEL_7`	Starting at Channel 7
`CHANNEL_9`	Starting at Channel 9
`CHANNEL_11`	Starting at Channel 11
`CHANNEL_13`	Starting at Channel 13
`CHANNEL_15`	Starting at Channel 15

14.2.14 TimeReference

Time Reference

Protocol used for time reference

enum

`NTP`	NTP (Network Time Protocol) time source
`PTP`	PTP (Precision Time Protocol) time source

14.2.15 VideoFormat

Format of the video essence.

struct

`resolutionScan`	**ResolutionScan** Resolution and scan mode of the source
`frameRate`	**FrameRate** Framerate of the source

15 hipValidate (1.0)

15.1 Type Reference

15.1.1 FloatRange

Float Range

struct

`min`	**float** Minimum value of the range
`max`	**float** Maximum value of the range

15.1.2 IntRange

Integer Range

struct

`min`	**int** Minimum value of the range
`max`	**int** Maximum value of the range

15.1.3 Result

Validation result

struct

`result`	**bool** True if config is valid and false otherwise.
`msg`	**optional string** An error message, which may be given if the result is invalid.

16 linkModes (1.3)

16.1 Type Reference

16.1.1 IpFecMode

enum

NO_FEC
RS_FEC
FC_FEC

16.1.2 IpLinkMode

enum

IP_10G
IP_25G_COPPER
IP_25G_OPTICAL
IP_40G
IP_4x10G

16.1.3 IpLinkSpeed

enum

IP_10G
IP_25G
IP_40G
IP_4x10G
NO_LINK

17 lldp (1.1)

17.1 Overview

```
# Changelog

## 1.1
- **Changed**
    - Extracted the structure inside of the `LldpNeighbor` object definition into a new type cal
```

17.2 Type Reference

17.2.1 LldpMedInventory

LLDP-MED Inventory. Appear specific implementation in parentheses.

struct

hwRev	**string** Hardware Revision
swRev	**string** Software Revision (Software Version)
fwRev	**string** Firmware Revision (FPGA Version)
serial	**string** Serial Number
manufacturer	**string** Manufacturer ("Appear")
model	**string** Model ("X10" or "X20")
assetId	**string** Asset ID (Hardware ID)

17.2.2 LldpMode

Choose either private or public configuration of LLDP frames.

enum

PUBLIC	Only mandatory TLVs and port detail TLVs.
PRIVATE	LLDP-MED inventory in addition to the PUBLIC TLVs.

17.2.3 LldpNeighbor

LldpNeighborStruct

17.2.4 LldpNeighborStruct

The optional variables are only present if GetLldpNeighborRequest.details is set to true in the rpc request. The following descriptions are the Appear specific implementations of each LLDP TLV. If neighbors from other manufacturers are detected, TLV values might differ. The LldpMode is specified in parentheses for variables that have different implementations based on the LldpMode set in the neighbor.

struct

`chassisId`	**string** MAC address of CTRL on MMI 1, fallback to CTRL on MMI 2 (PRIVATE) or "N/A" (PUBLIC).
`mgmtIp`	**list** of **string** IPv4 address of CTRL on MMI 1 and/or CTRL on MMI 2. Also IPv6 if enabled on the neighbor.
`sysDescr`	**optional string** Desktop Heading (PRIVATE) or "Unknown" (PUBLIC)
`sysName`	**optional string** Hostname (PRIVATE) or Interface.Port (PUBLIC)
`portId`	**string** MAC address of local port
`portDescr`	**optional string** Slot, Interface: Port-label
`ttl`	**optional string** Time-To-Live, how long an LLDP packet received from that neighbor should be considered valid.
`lldpMedInventory`	**optional LldpMedInventory** LLDP-MED Inventory

18 lldpProtocols (1.0)

19 networkManagerProtocols (1.1)

19.1 Overview

Changelog:

1.1

- **Changed**
  - The imported package `hipIpInterface` is bumped from 1.1 to 1.2.

1.0

- copied from firewall networkManagerProtocol.1.7

20 nmosConfig (1.1)

20.1 Overview

```
# Changelog

## 1.1
- **Added**
    - Added new type `ConnectionType`
    - Added member 'connectionType' to NmosConfig
```

20.2 Type Reference

20.2.1 ConnectionType

Enum to control whether the NMOS traffic is in-band (through line card dataports) or out-of-band (through the MMI).

enum

IN_BAND	
OUT_OF_BAND	

20.2.2 NmosConfig

Configuration parameters for setting up NMOS.

struct

connectionType	**ConnectionType** Configuration to control whether traffic should be in-band or out-of-band
registry	**optional nmosRegistryConfig.NmosRegistryConfigStruct** Configuration parameters for connecting to the NMOS registry
labelTemplates	**optional nmosLabelConfig.NmosLabelConfig** Custom NMOS labels configuration. This feature is enabled when the optional is set

21 nmosLabelConfig (1.0)

21.1 Overview

Changelog

1.0

NMOS labelling

NMOS labelling API allows operator to define templates based on auto text options. This templates can be defined for for node, device, source, flow, sender and receiver.

Version 1.0 allows six auto text options.

What are the auto text options?

- ${SLOT} - The slot number of the card in use.
- ${DEV_ID} - The Device number. **Not supported for Node templates**.
- ${ESS_GR} - The label of the group where the essences belong to. **Not supported for Node templates**.
- ${ESS} - The label of the Essence in question. **Not supported for Node and Device templates**.
- ${ESS_T} - The Essence type: Video, Audio, Ancillary. This auto text option is substituted by what is defined in `EssenceTypeStrings` for each of the essences type. **Not supported for Node and Device templates**.
- ${ESS_T_INDEX} - The number of the essence belonging to one of the essence type - video, audio, ancillary. **Not supported for Node and Device templates**.

How to use the NMOS Labelling.

- NMOS templates are set in the fields part of `ResourceLabelTemplates`. Unused templates should be set with an empty string.
- Each field in `ResourceLabelTemplates` can have as many auto text options as the operator wants, as long as they are supported for the respective template.
- `EssenceTypeStrings` strings will substitute the auto text option ${ESS_T} with the defined string for the respective Essence type.

Example of strings to use for the fields in `ResourceLabelTemplates` and `EssenceTypeStrings`

- `nodeTemplate` - ECx210 Decoder ST2110 Sender Slot ${SLOT}
- `deviceTemplate` - ECx210 Decoder Sender Slot ${SLOT} ID ${DEV_ID}
- `sourceTemplate` - ECx210 Slot ${SLOT} ID ${DEV_ID}, ${ESS_T} Source: ${ESS}, Group: ${ESS_GR}, Essence Number: ${ESS_T_INDEX}
- `flowTemplate` - ECx210 Slot ${SLOT} ID ${DEV_ID}, ${ESS_T} Flow: ${ESS}, Group: ${ESS_GR}, Essence Number: ${ESS_T_INDEX}
- `senderTemplate` - ECx210 Slot ${SLOT} ID ${DEV_ID}, ${ESS_T} Sender: ${ESS}, Group: ${ESS_GR}, Essence Number: ${ESS_T_INDEX}
- `receiverTemplate` - ECx210 Slot ${SLOT} ID ${DEV_ID}, ${ESS_T} Receiver: ${ESS}, Group: ${ESS_GR}, Essence Number: ${ESS_T_INDEX}
- `video` - Video
- `audio` - Audio
- `ancillary` - Anc

21.2 Type Reference

21.2.1 EssenceTypeStrings

Labels for each essence type to use when generating custom NMOS labels.

struct

`video`	**string** Video essence label
`audio`	**string** Audio essence label
`ancillary`	**string** Ancillary essence label

21.2.2 NmosLabelConfig

Custom NMOS labels configuration.

struct

`resourceTemplates`	**ResourceLabelTemplates** Templates for each NMOS resource
`essenceStrings`	**EssenceTypeStrings** Labels to use for each essence type

21.2.3 ResourceLabelTemplates

NMOS label templates for each NMOS resource.

struct

`nodeTemplate`	**string** Template string for the node label
`deviceTemplate`	**string** Template string for the device label
`sourceTemplate`	**string** Template string for the source label
`flowTemplate`	**string** Template string for the flow label
`senderTemplate`	**string** Template string for the sender label
`receiverTemplate`	**string** Template string for the receiver label

22 nmosRegistryConfig (1.0)

22.1 Type Reference

22.1.1 DnsConfig

Method to use for discovery of registry server

variant

`mDns`	**DnsConfig.mDns** Use mDNS to reach the registry server
`dnsSd`	**DnsSdConfig** Configuration parameters to use DNS-SD for discovery of registry service

22.1.2 DnsConfig.mDns

empty **struct**

22.1.3 DnsSdConfig

Configuration parameters for DNS-SD

struct

`dnsServer`	**firewallTypes.IpAddress** IPv4 address of DNS-SD server
`search`	**string** Search domain name

22.1.4 NmosRegistryConfigStruct

Configuration parameters for reaching the NMOS registry

struct

`registryConfig`	**RegistryManualConfig** Configuration parameters to use in case of failure in mDNS or DNS-SD (secondary/fallback method)
`dnsConfig`	**DnsConfig** Configuration parameters for mDNS or DNS-SD (primary method)

22.1.5 RegistryManualConfig

Registry information to be used in case registry server can not be reached by mDNS or DNS-SD.

struct

`fallbackEnable`	**bool** Enable use of given information as a fallback
	firewallTypes.IpAddress

`registryIpAddr`	IPv4 address of NMOS registry
`registryPort`	**int** Port number of NMOS registry
`registryVersion`	**commonTypes.NmosVersionNumber** Version of registry API

23 nmosRegistryStatus (1.0)

23.1 Type Reference

23.1.1 NmosRegistryStatus

NMOS registry status

struct

`ipAddr`	**optional firewallTypes.SocketAddress** Currently used IP address and port number of the registry server. This value may have been obtained from DNS-SD, mDNS, or a manual IP entry.
`registered`	**bool** True if there is an active connection to the registry server.

24 nmosStatus (1.1)

24.1 Overview

Changelog:

1.1

```
- **Added**
    - The field 'nodeApi' was added to 'NmosStatus'.
- **Changed**
    - The type NmosStatus was moved to the file 'nmosStatusTypes'.
```

24.2 Command Reference

24.2.1 GetNmosStatus

RPC for requesting NMOS status

- **message GetNmosStatus.Request**
- **message GetNmosStatus.Response**

24.3 Type Reference

24.3.1 GetNmosStatus.Request

empty **struct**

24.3.2 GetNmosStatus.Response

nmosStatusTypes.NmosStatus

25 nmosStatusTypes (1.0)

25.1 Type Reference

25.1.1 NmosStatus

NmosStatus Status of the NMOS service

struct

`nmosRegistryStatus`	**nmosRegistryStatus.NmosRegistryStatus** Status of the NMOS registry connection
`nodeApi`	**map** from **physicalIpPort.PortName** to **string** Map from port to the endpoint for NMOS Node API. Ex: PortName.D1 -> “10.20.10.34:8001”, PortName.D2_2 -> “10.20.11.34:8001”

26 physicalIpPort (1.0)

26.1 Type Reference

26.1.1 PortName

enum

CTRL	
D1	
D2	
D3	
D4	
D1_1	
D1_2	
D1_3	
D1_4	
D2_1	
D2_2	
D2_3	
D2_4	
D3_1	
D3_2	
D3_3	
D3_4	
D4_1	
D4_2	
D4_3	
D4_4	

27 sfpConstants (1.3)

27.1 Type Reference

27.1.1 Identifier

Physical Device Identifier Values. (SFF 8472 Rev12.3 spec Table 5-1)

enum

UNKNOWN	Unknown or unspecified
GBIC	GBIC
SOLDERED	Module soldered to motherboard (ex. SFF)
SFP_SFPP	SFP or SFP+
QSFPP	QSFP+
QSFP28	QSFP28
NOT_USED	Not used by SFF 8472 Rev12.2.1. These values are maintained in the Transceiver Management section of SFF-8024.
VENDOR	Vendor specific

27.1.2 Transceiver

Transceiver Compliance Codes. (SFF 8472 Rev12.3 spec Table 5-3)

enum

ER_10G	10G Base-ER
LRM_10G	10G Base-LRM
LR_10G	10G Base-LR
SR_10G	10G Base-SR
ACTIVE_CABLE_40G	40G Active cable
LR4_40G	40G Base-LR4
SR4_40G	40G Base-SR4
CR4_40G	40G Base-CR4
PX	BASE-PX
BX10	BASE-BX10
FX_100	100BASE-FX
LX_100	100BASE-LX/LX10
T_1000	1000BASE-T
CX_1000	1000BASE-CX
LX_1000	1000BASE-LX
SX_1000	1000BASE-SX
ELECTRICAL_INTER_ENCLOSURE	Electrical inter enclosure
LONGWAVE_LASER_LL	Longwave laser LL
MEDIUM	Medium distance
LONG_DISTANCE	Long distance
INTERMEDIATE_DISATANCE	Intermediate distance
SHORT_DISTANCE	Short distance
VERY_LONG_DISTANCE	Very long distance
ACTIVE_CABLE	Active Cable
PASSIVE_CABLE	Passive Cable

LONGWAVE_LASER_LC	Longwave laser LC
SHORTWAVE_LASER_w_OFC	Shortwave laser w OFC
SHORTWAVE_LASER_wo_OFC	Shortwave laser w/o OFC
ELECTRICAL_INTRA_ENCLOSURE	Electrical intra enclosure
SINGLE_MODE	Singel mode
MULTIMODE_50um	Multimode 50um
MULTIMODE_50m	Multimode 50m
MULTIMODE_625m	Multimode 62.6m
VIDEO_COAX	Video coax
MINIATURE_COAX	Miniature coax
SHIELDED_TWISTED_PAIR	Shielded twisted pair
TWIN_AXIAL_PAIR	Twin axial pair
M_BYTES_SEC_100	100M bytes per sec
M_BYTES_SEC_200	200M bytes per sec
M_BYTES_SEC_400	400M bytes per sec
M_BYTES_SEC_1600	1600M bytes per sec
M_BYTES_SEC_800	800M bytes per sec
M_BYTES_SEC_1200	1200M bytes per sec Extended Specification Compliance. (SFF-8024 Rev 4.8 spec Table 4-4)
AOC_100G_AOC_25GAUI_C2M_BER_5_	100G AOC (Active Optical Cable) or 25GAUI C2M AOC. Providing a worst BER of $5 * 10^{-5}$
SR4_100G_BASE_SR_25G_BASE	100GBASE-SR4 or 25GBASE-SR
LR4_100G_BASE_LR_25G_BASE	100GBASE-LR4 or 25GBASE-LR
ER4_100G_BASE_ER_25G_BASE	100GBASE-ER4 or 25GBASE-ER
SR10_100G_BASE	100GBASE-SR10
CWDM4_100G	100G CWDM4
PSM4_PARALLEL_SMF_100G	100G PSM4 Parallel SMF
ACC_100G_ACC_25GAUI_C2M_BER_5_	100G ACC (Active Copper Cable) or 25GAUI C2M ACC. Providing a worst BER of $5 * 10^{-5}$
CR4_100G_BASE_25G_BASE_CR_CA_2	100GBASE-CR4 or 25GBASE-CR CA-L
CR_25G_BASE_CA_25G_S_50G_BASE_	25GBASE-CR CA-S
CR_25G_BASE_CA_25G_N_50G_BASE_	25GBASE-CR CA-N
ER4_40G_BASE	40GBASE-ER4
SR_4_X_10G_BASE	4 x 10GBASE-SR
PSM4_PARALLEL_SMF_40G	40G PSM4 Parallel SMF
G959_1_PROFILE_P1I1_2D1	G959.1 profile P1I1-2D1 (10709 MBd, 2km, 1310nm SM)
G959_1_PROFILE_P1S1_2D2	G959.1 profile P1S1-2D2 (10709 MBd, 40km, 1550nm SM)
G959_1_PROFILE_P1L1_2D2	G959.1 profile P1L1-2D2 (10709 MBd, 80km, 1550nm SM)
T_10G_BASE	10GBASE-T with SFI electrical interface
CLR4_100G	100G CLR4
AOC_100G_AOC_25GAUI_C2M_BER_10	100G AOC or 25GAUI C2M AOC. Providing a worst BER of 10^{-12} or below
ACC_100G_ACC_25GAUI_C2M_BER_10	100G ACC or 25GAUI C2M ACC. Providing a worst BER of 10^{-12} or below
DWDM2_100GE	100GE-DWDM2 (DWDM transceiver using 2 wavelengths on a 1550nm DWDM grid with a reach up to 80km)

28 sfpStatus (1.4)

28.1 Type Reference

28.1.1 SfpDiagnostics

Real Time Diagnostic for SFP/SFP+/QSFP28/QSFP+, with A2h/A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 9-11) (SFF 8636 Rev2.10 spec Table 6-1)

struct

`temp`	**float** Internally measured module temperature (°C). [96-97], [22-23]
`vcc`	**float** Internally measured supply voltage in transceiver (V). [98-99], [26-27]
`txPwr`	**list** of **float** Measured TX output power (mW). [102-103], [50-57]
`rxPwr`	**list** of **float** Measured RX input power (mW). [104-105], [34-41]

28.1.2 SfpStatus

SFP Status Data Fields, with A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 4-1) QSFP A0h memory space address is specified next to the SFP A0h. (SFF 8636 Rev 2.10 spec Table 6-14)

struct

`identifier`	**sfpConstants.Identifier** Type of transceiver (SFP/SFP+ or unknown). [0],[128]
`transceiver`	**list** of **sfpConstants.Transceiver** Code for electronic or optical compatibility [3-10], [131-138]
`brNominal`	**float** Nominal signalling rate, units of 100 MBd. [12], [140/222]
`vendorName`	**string** SFP vendor name. [20-35], [148-163]
`vendorOUI`	**string** SFP vendor IEEE company ID. [37-39], [165-167]
`vendorPN`	**string** Part number provided by SFP vendor. [40-55], [168-183]
`vendorRev`	**string** Revision level for part number provided by vendor. [56-59], [184-185]
`wavelength`	**string** Laser wavelength. [60-61], [186-187]
`vendorSN`	**string** Serial number provided by SFP vendor. [68-83], [196-211]
`dateCode`	**string** Vendor's manufacturing date code. [84-91], [212-219]
	optional SfpDiagnostics

`diagnostics`	Optional Diagnostics Monitor Data.

29 st2110Types (1.1)

29.1 Type Reference

29.1.1 St2110Standard

The SMPTE ST2110 standards that describe video, audio or ancillary content.

enum

ST2110_20	ST2110 uncompressed video standard
ST2110_22	ST2110 compressed video standard
ST2110_30	ST2110 audio standard
ST2110_31	ST2110 AES3 formatted audio
ST2110_40	ST2110 ancillary standard